

```

124. # Calculate performance ratios
125. if len(results_df) > 0:
126.     results_df['Sharpe Retention'] = results_df['Test Sharpe'] / results_df['Train Sharpe']
127.
128. # Calculate average metrics
129. avg_train_sharpe = results_df['Train Sharpe'].mean()
130. avg_test_sharpe = results_df['Test Sharpe'].mean()
131. avg_retention = results_df['Sharpe Retention'].mean()
132.
133. print("\nWalk-Forward Validation Summary:")
134. print(f"Average Training Sharpe: {avg_train_sharpe:.4f}")
135. print(f"Average Testing Sharpe: {avg_test_sharpe:.4f}")
136. print(f"Average Sharpe Retention: {avg_retention:.2%}")
137.
138. return results_df
139.

```

The key insight from walk-forward validation is the "Sharpe Retention" ratio—how much of the strategy's risk-adjusted performance carries over from the training period to the out-of-sample testing period. A consistently high retention ratio (ideally above 70%) suggests the strategy is robust and not overfit.

*?? **Did You Know?** In 2007, several prominent quant hedge funds, including Renaissance Technologies and D.E. Shaw, experienced unprecedented losses in what became known as the "Quant Quake." An analysis later revealed that many firms had inadvertently developed similar strategies, all optimized on the same historical data. When one large fund began unwinding their positions, it created a cascade effect that punished these similar strategies. This crisis highlighted a hidden risk of optimization: if everyone optimizes on the same data, they might converge on similar strategies, creating systemic vulnerabilities. Since then, sophisticated quant firms have incorporated "uniqueness" as an explicit objective in their strategy development process, actively avoiding approaches that resemble common industry patterns—even if they appear optimal in standard backtests.*